

PRAGYA — Plan 1: Team & Workload

Project: PRAGYA — FLN Learning Gap Agent (PS32, EduTech track). This plan locks the concept exactly as finalized: one vertical (subtraction + regrouping), the 7-stage pipeline, and the role split below — no scope drift from here. Hack-e-Awadh · Saturday 26 September 2026 · Hotel Charans Plaza, Hazratganj.

"Don't just find the gap. Find the gap behind the gap."

The Locked Concept

PRAGYA is a teacher-facing diagnostic tool. Instead of "this child is weak in subtraction," it finds *why* — verifies the cause with 3 short questions, groups students by shared root cause, recommends one activity per group, and re-tests to confirm the gap closed.

Pipeline: Assess → Trace → Diagnose → Verify → Group → Intervene → Reassess.

Locked example: item 3 of the 5-item assessment, $83 - 47 = 46$ ❌. Candidate causes: calculation slip, place-value confusion, regrouping misconception, word-problem difficulty. Three micro-diagnostic questions narrow it to one, with evidence. Verified: regrouping.

One rule that never changes: rules decide *what* happened (scoring, skill mapping); AI only helps investigate *why*, and never picks the final diagnosis alone — the teacher can always accept, reject or change it.

Constraints

Constraint	What it means for us
6 build hours: 10:00–13:00, 13:30–16:30	The schedule below follows the hackathon's own session times
All 3 vibe-code with AI agents — Antigravity (all three, unlimited), Claude Code (Abhinav only, limited)	Every module is agent-built; humans instruct, review and test
Pencils down 16:30, hard	Feature freeze 15:15, code freeze 15:45
Fresh code only, public repo	Repo created at 10:00. Tonight = content and prompts, no code

Constraint	What it means for us
All 3 present, each presents their own part	Each role below has a distinct, presentable module
One pipeline, not three mini-projects	Modules hand off in a fixed order (below); everyone can trace a diagnosis end to end

Roles (locked as in the main idea)

Person	Role	Owns	Says to judges
Abhay	Intelligence & Integration Lead	Root-cause engine, cause graph, diagnostic (verify) decision logic, scoring, final system integration	"I built PRAGYA's intelligence layer — the pipeline that turns an observed error into competing root-cause hypotheses and verifies them through targeted micro-diagnostics."
Ankit	Response Processing & Assessment Engineer	Response input, question/answer structure, error-identification screen, (P1) image upload	"I built the assessment layer that turns a student's raw response into structured evidence PRAGYA can analyze."
Abhinav	Teacher Dashboard & Intervention Engineer	Teacher dashboard, verified-gap view, student grouping, intervention screen, reassessment + before/after; also owns the shared data contract and a cross-repo review pass	"I built the teacher-facing layer — verified gaps become student groups, targeted interventions, and measurable reassessment results."

The hand-off, one pipeline: Ankit → Student Response → Abhay → Root-Cause Diagnosis → Abhinav → Intervention + Group → Abhay → Reassessment.

Why Abhinav also owns the contract and review pass: only his machine has Claude Code. He writes `lib/types.ts` at 10:00 so all three modules agree on the data shapes before building, then runs one Claude Code review of the whole repo at the 12:30 and 15:15 checkpoints — catching bugs across all three modules, not just his own.

Feature List

P0 — the locked concept, must work by feature freeze (15:15). This is the full main idea's scope: subtraction + regrouping only.

#	Feature	Owner	Done when
1	Class/Student profile: teacher creates a class and adds a student list (or loads the seeded demo class)	Ankit	A class with named students exists before any assessment runs
2	Cause graph: place value, regrouping, calculation slip, word problem — each with its evidence rule	Abhay	Loads with all 4 causes and their rules
3	Assess: teacher runs the short 5-item predefined assessment for a student (item 3 is the locked 83 – 47 example)	Ankit	All 5 responses saved as structured data
4	Trace + Diagnose: candidate causes ranked from which items were missed, across all 5	Abhay	Returns 2–4 ranked candidates, each with a reason
5	Verify: the 3 locked micro-diagnostics — "value of 8 in 83?", "subtract 7 from 3 directly?", "show how you'd regroup 83"	Abhay logic, Ankit UI	Verdict shown as a count ("2 of 3 point to regrouping"), never a guessed percentage
6	Teacher Accept / Reject / Change on the verified gap	Ankit	Teacher's decision is the final record
7	Class run: the same 5-item assessment for a seeded class of 30; Group stage clusters by verified cause	Abhinav	Reproduces the locked split: 10 regrouping, 5 place value, 2 calculation, 1 word problem, 12 correct
8	Intervene: one activity per group	Abhay content, Abhinav UI	Each group shows its own activity
9	Reassess: 3 fresh items per group; before/after shown as a real percentage computed from actual correct counts	Abhay logic, Abhinav UI	Gap-closure number traces to real responses, never invented

#	Feature	Owner	Done when
10	Visual gap report: class heat-map / bottleneck bars	Abhinav	Matches Plan 2's classroom bottleneck numbers
11	Live public URL	Abhinav	Opens on a phone

P1 — our additions, built only in slack time, after every P0 checkpoint above is met. Never risk a P0 checkpoint to start these.

#	Feature	Slot
12	Addition-with-carry as a second skill, same pipeline	Only if the 12:30 integration checkpoint lands early
13	AI-written teacher explanation + Hindi translation of the activity (cached fallback)	13:30–14:30 slack
14	Printable blank assessment sheet (the 5 items, for paper-and-pencil administration in a classroom without devices)	13:30–14:30 slack
15	Printable results sheet (the class run + the group's activity)	14:30–15:15 slack
16	Parent report per child: which skill they're working on, one home activity, English/Hindi, printable	14:30–15:15 slack
17	Photo upload of a handwritten answer, teacher confirms the reading before it's used	Only if everything above is solid by 14:30
18	Landing page, saved/exported data, full language toggle, mobile polish	Only if feature freeze (15:15) arrives early

P2 — attempt only if every P1 item above is done and time still remains before 15:15. If not, these go on the roadmap slide, and that's a fine outcome.

#	Feature	Why it's P2, not P1
19	Voice-based assessment: child speaks a number, browser speech-to-text reads it, teacher confirms the digit before it's recorded (never auto-saved)	Reading a spoken number is far safer than open transcription, but still needs real testing time in a noisy room — it must not be the first time it's tried live
20	Read-aloud, WhatsApp share, question builder, installable app, login	Roadmap slide — say so if asked, don't attempt tomorrow

Schedule (the hackathon's own session times)

Time	Abhay	Ankit	Abhinav
08:30–09:30	Ask organisers: agents allowed as described? Submission format? Meet the mentor	Wi-Fi, hotspot, Antigravity logged in	Wi-Fi, hotspot, Antigravity + Claude Code logged in
09:30–10:00	Briefing	Briefing	Briefing
10:00–10:15	All 3 + mentor: walk through the locked scope, architecture, roles; ask the mentor what could undermine the diagnosis's credibility		Writes <code>lib/types.ts</code> , pushes first commit
10:15–12:30	Agent-builds the cause graph + diagnose logic across all 5 items (features 2, 4)	Agent-builds the class/student profile screen and the 5-item assess screen (features 1, 3)	Agent-builds the dashboard shell + verify/verdict UI (features 5, 6)

Time	Abhay	Ankit	Abhinav
12:30–13:00	First integration, all 3: Input → Diagnose → Verified gap works for one student, end to end. Claude Code review pass #1 (Abhinav)		
13:00–13:30	Lunch	Lunch	Lunch
13:30–14:30	Wires Verify's 3-question flow fully; starts Reassess logic (feature 8)	Builds Accept/Reject/Change (feature 5) and the class-run input (feature 6, UI half)	Builds Group + Intervene screens (features 6, 7)
14:30 checkpoint	Diagnose → Verify → Intervene works for one student		
14:30–15:15	Finishes Reassess logic; P1 items if ahead	Class-run + grouping fully wired; P1 items if ahead	Deploys; P1 items if ahead; Claude Code review pass #2
15:15 Feature freeze	Group → Reassess → Gap closure works for the seeded class of 30		
15:15–15:45	Bug fixes only	Bug fixes only	Bug fixes only; final deploy
15:45 Code freeze			
15:45–16:05	Full rehearsal, all 3, timed		
16:05–16:20	Submission form, repo link, backup video		
16:20–16:30	Buffer. Pencils down		

Fallback Ladder

Trigger	What we do
12:30 — first integration not working	Stop all new work; all 3 fix it together; every P1 item is cancelled
14:30 — Verify/Intervene not working	Drop the class-run (features 6–8) to a single scripted student; demo that instead
Agent stuck or breaks working code	Revert to the last commit; write a smaller instruction
An LLM call (Hindi/explanation) is slow or down	Serve cached text — the demo never waits on an API
15:15 — anything still broken	Hide it. Never demo a half-working screen
Laptop dies	Second laptop has the repo; the live URL and a backup video exist

Tonight (no code)

- **Abhay:** hand-check the exact wrong-answer values for all 5 assessment items (the locked example is item 3) and the class-run split (Plan 2, sections 3, 5, 7, 14), so the numbers are right before any agent touches them. Prepare the mentor line and the closing line (Plan 2, sections 16, 15).
- **Ankit:** sketch the assess screen and the verify (3-question) screen on paper.
- **Abhinav:** save `AGENTS.md` and the prompt pack (Plan 2, section 19) in a notes file; get a separate LLM API key for the app's own AI calls — not the same login as Antigravity or Claude Code.
- **Everyone:** charge laptops and a power bank, set up a phone hotspot, pack an umbrella, arrive by 08:15.

Presenting — 90 Seconds Locked, up to 3:00 if Time Allows

Time	Speaker	Screen
0:00–0:15	Abhay	The problem: $83 - 47 = 46$ ❌ — "weak in subtraction" is the wrong diagnosis

Time	Speaker	Screen
0:15–0:35	Ankit	Assess: enters the response; system detects a subtraction error and lists candidate causes
0:35–0:55	Abhay	Verify: the 3 micro-diagnostics run live; verified root gap: regrouping, with evidence
0:55–1:15	Abhinav	Group: 10 of 30 students share this exact gap; Intervene: one targeted activity for the group
1:15–1:30	Abhinav	Reassess: gap-closure number, computed from real responses
1:30–1:45 (if time)	Abhay	"A wrong answer is a symptom. PRAGYA finds the underlying skill gap, verifies it, and turns that diagnosis into action."